USB CABLE PLUG-IN GUIDE · PHYSICAL PORTS

Where to plug each camera

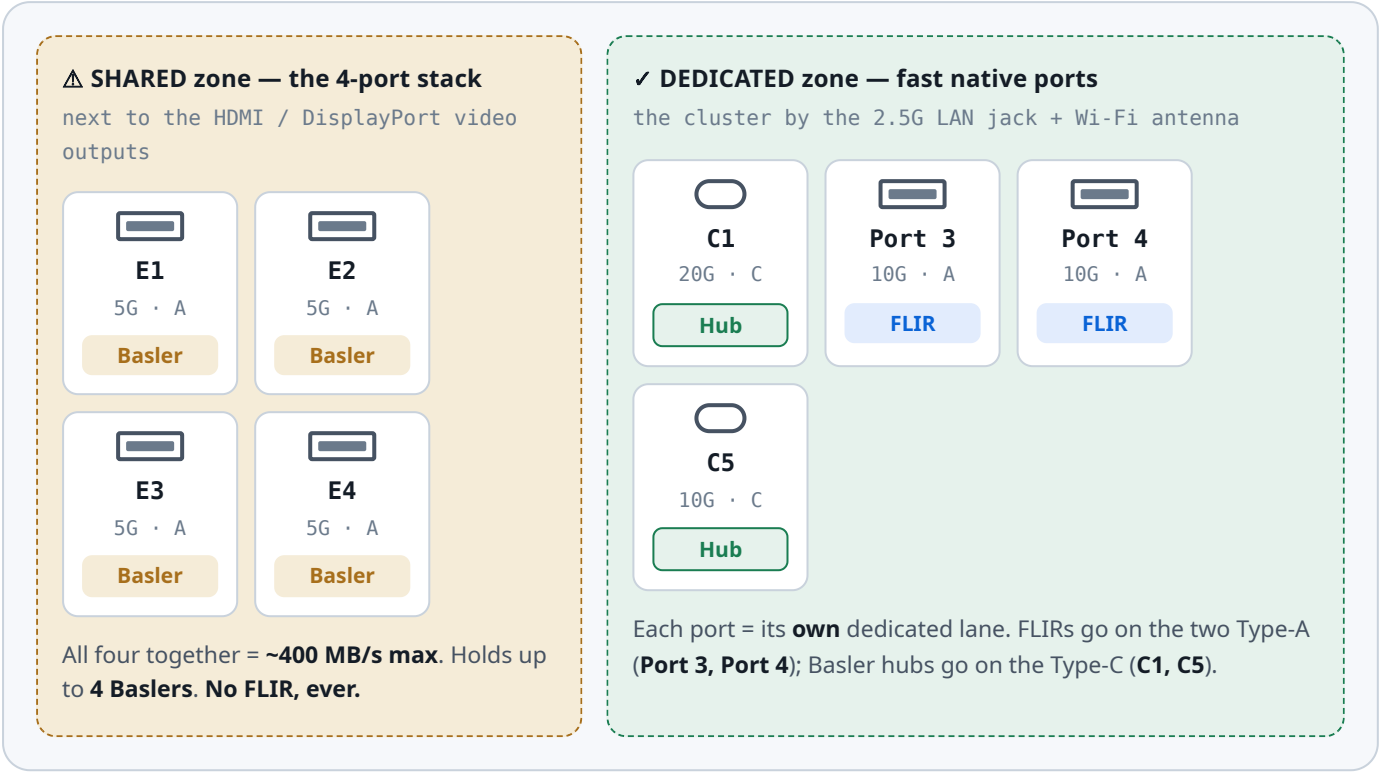
A cable-by-cable guide to the rear and front panels for 2× FLIR + 8× Basler. The panel has two USB zones — one is a shared hub, one is dedicated lanes. Getting the FLIRs into the right zone is the whole game.

ASUS PRIME Z790-A WIFI · rear-I/O labels per user manual p.2-21

⚡ **The one rule:**
The **block of 4 identical stacked Type-A ports next to the HDMI / DisplayPort outputs** all share ONE 5 Gbps pipe (they're behind an onboard hub). Put **Baslers only** there — **never a FLIR**. Each FLIR needs its own port in the other zone.

01 The rear panel, by zone

You don't need to read the tiny silk-screen labels — the two zones are told apart by what's next to them.



Connector legend: ▭ = Type-A (rectangular, what your camera cables plug straight into) · ○ = Type-C (oval, reversible — used for the Basler hubs' upstream cable).

02 Cable-by-cable

Both camera models present a **USB Type-A plug on the host end**, so they drop straight into any Type-A port. The Baslers ride 4 to a hub; the hub's *upstream* cable is Type-C into C1 / C5.

CABLE	PLUG INTO	WHERE IT IS	WHY
FLIR **#1**	rear **Port 3** (Type-A, 10G)	LAN / Wi-Fi cluster	own dedicated lane — a FLIR nearly fills a 5G link by itself
FLIR **#2**	rear **Port 4** (Type-A, 10G)	LAN / Wi-Fi cluster	own dedicated lane, separate from FLIR #1
BASLER **×4**	**Hub #1** → its 4 Type-A ports	hub sits on desk	4 Baslers ≈ 325 MB/s — fits a 10G hub with room to spare
Hub #1 uplink	rear **C1** (Type-C, 20G)	LAN / Wi-Fi cluster	gives the 4 Baslers a ~900 MB/s shared pipe
BASLER **×4**	**Hub #2** → its 4 Type-A ports	hub sits on desk	the other 4 Baslers
Hub #2 uplink	rear **C5** (Type-C, 10G)	LAN / Wi-Fi cluster	~900 MB/s shared pipe, independent of Hub #1

🔌 **Hub spec that matters:**
Buy **powered** (self-powered) **USB 3.2 Gen 2 (10 Gbps)** hubs with a **Type-C upstream**. A cheap "USB 3.0 / 5G" hub — or reusing the board's own 4-port stack — caps all its cameras at ~400 MB/s and defeats the point. The 10G upstream is where the extra headroom comes from.

03 Two simpler variants

Running only 6 cameras? Skip the hubs.

2 FLIR → **Port 3 & Port 4**. 4 Basler → the **E1–E4 stack** (that's exactly your current wiring — it works, zero drops). Only add hubs when you go past 4 Baslers.

One hub instead of two.

All 8 Baslers on a single powered **7–10-port Gen 2** hub → **C1** (20G). That's ~650 / 900 MB/s ≈ 72% — fine. Two hubs (4+4) just leaves more margin and is easier to cable-manage.

Front panel: the front Type-C connector (10G) is also a dedicated lane and is a fine spot for one Basler or a hub if the rear is crowded. Leave the front USB 2.0 header ports for the keyboard / mouse / trigger box — they're too slow for cameras.

04 Confirm it in 2 minutes

The port *types* and the 4-stack are unmistakable by eye, but if you want to be certain which physical socket is which, plug one camera in at a time and read the kernel — a **bare** number (`2-6`) means a dedicated lane; a **dotted** one (`2-9.3`) means it landed on the shared hub.

```
for d in /sys/bus/usb/devices/*/; do [ -f "$d/devpath" ] || continue
  printf '%s  devpath=%s speed=%s  %s\n' "$(basename $d)" \
    "$(cat $d/devpath)" "$(cat $d/speed)" "$(cat $d/product 2>/dev/null)"
done | sort -V
```

✓ **Good:** each FLIR shows a bare devpath like `2-3` at `speed=5000` — its own lane. (5000 is normal; the cameras are 5G devices even on a 10G port.)

✕ **Fix it:** a FLIR showing a dotted devpath `2-9.x` is on the shared hub — move it to a Port 3 / Port 4 socket.

3 Then run a full capture and check `recording_summary.json`: any camera short of `round(fps×duration)` frames is bandwidth-starved.

Confidence: the 4-stack = the shared 5G hub is proven from your live system; exact silk-screen labels (E1–E4 / C1 / Port 3–4 / C5) follow the ASUS manual p.2-21. If any doubt on which socket is which, the 2-minute check above is definitive. · Full bandwidth analysis: see the companion topology report.